Autonomous Retail.

AI-CM is an agentic co-pilot for Category Managers, replacing manual data hunting with a self-correcting workforce on AWS Graviton2 and Bedrock.

1. Why AI is Required

The Linguistic Barrier

Rule-based NLP requires rigid dictionary mapping. AI handles spelling, jargon, and intent weight instinctively, ensuring managers don't need technical query knowledge.

AI Advantage: Native semantic intent isolation handles spelling and business jargon instinctively.

Dynamic SQL Self-Healing

Mapping to 15+ tables via templates is brittle. Our Analyst agent uses a ReAct loop to interpret schema errors and self-correct SQL on-the-fly without developer intervention.

AI Advantage: Autonomous navigation of complex star-schema relational data.

2. The Cognitive Workforce

Supervisor

ORCHESTRATOR

ARCHITECTURE

Few-shot classification with a **Clarification Gate** to handle vague queries at zero token cost.

AI NECESSITY

Isolates intent in multi-context strings where keyword matchers fail.

Analyst

TEXT-TO-SQL

ARCHITECTURE

Implements **ReAct reasoning** loops to interpret DB errors and self-heal SQL syntax dynamically.

AI NECESSITY

Navigates 15+ relational tables without needing brittle manual templates.

Strategist

REASONING

ARCHITECTURE

Uses **pgvector Semantic Index** to inject relevant historical context via RAG for consistency.

AI NECESSITY

Evaluates causal links between stock and pricing trends that rules can't model.

Planner

ACTION ENGINE

ARCHITECTURE

Outputs strict JSON for `action_log`. Maps fuzzy strategy to deterministic execution steps.

AI NECESSITY

Assigns contextual confidence scores for automated execution proposals.

Liaison

STAKEHOLDER COMMS

ARCHITECTURE

Drafts context-aware professional stakeholder emails using RAG tone personalization.

AI NECESSITY

Replaces robotic templates with tactful, situational business prose.

Watchdog

MONITORING

ARCHITECTURE

A robust hybrid system performing 95% of checks via SQL-based thresholding through a **Distributed Cron scheduler (pgxpool)**. It only escalates to Bedrock LLM for qualitative interpretation of non-linear data drift.

AI NECESSITY

Reduces alert fatigue by validating qualitative anomalies that deterministic rules flag as noise, ensuring high-signal notifications.

Recommender

RULE ENGINE

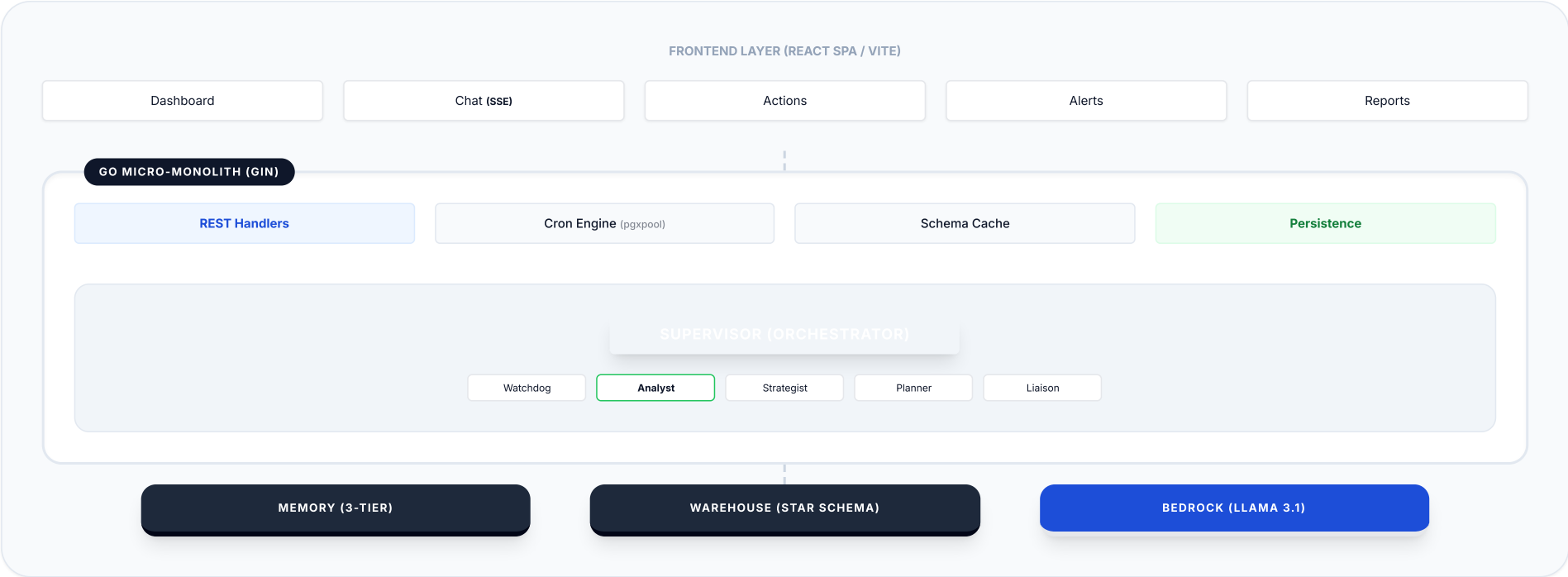
ARCHITECTURE

A specialized **Go-native heuristics engine** executing pre-optimized analytical SQL against the relational warehouse. Engineered for sub-second latency to support rapid lookups.

ARCHITECTURE ROLE

Governs deterministic business logic such as inventory reorder levels and price matching, preserving expensive LLM tokens for strategy layers.

3. System Logic Architecture



4. Cloud & CI/CD Architecture

5. AI Layer: UX Value

Frictionless Discovery

Natural language eliminates technical barriers. Users **"Ask"** and get complex metrics immediately without knowing SQL or BI filters.

Causal Clarity

AI connects pricing shifts to stockouts instinctively, providing the **"Why"** narrative behind trends for consistent strategic context.

Proactive Guardrails

The Watchdog agent monitors data 24/7, catching anomalies before they hit quarterly margins, moving users from reactive to proactive monitoring.

Actionable Results

Planner converts fuzzy goals into ready-to-execute price changes or restock orders in one single click via the integrated Action Center.

Human-in-the-Loop Workflow

- ✓ **Deep Memory:** AI remembers previous strategies and stakeholder preferences using pgvector to avoid redundancy and maintain continuity.
- ✓ **Process Clarity:** Real-time SSE streaming shows the agent's thought process as it happens, building trust through total procedural transparency.

NATURAL QUERY	USER INPUT
INTENT CONTEXT	SUPERVISOR
HEURISTIC SQL	ANALYST
PROPOSED PLAN	PLANNER

6. AWS Infrastructure Stack

COMPUTE

EC2 t4g.small

Graviton2 ARM64 provides **2GB RAM**—optimized for high-density Docker monolith stability while qualifying for the promotional free trial.

DATABASE

RDS PostgreSQL

Managed instance with `pgvector`. Handles relational sales facts and semantic memory search simultaneously in a single managed tier.

INFERENCE

Bedrock

Serverless **Llama 3.1 70B** in us-east-1, delivering high-tier reasoning at **14x lower cost** than proprietary API alternatives.